



CYBER ATTACK ANALYSIS



Global Large Bank - Critical Incident Response



SCENARIO 1: RANSOMWARE LOCKOUT

All victim files locked including central data repositories, servers, and workstations



SCENARIO 2: DATA EXTORTION

Cyber gang claims data theft, threatens release, posts victim on dark web with data samples



Analysis Questions



Detailed Response

a. Type of Cyber Attack

DOUBLE EXTORTION RANSOMWARE ATTACK

Scenario 1 + 2 Combined: This sophisticated attack combines traditional ransomware (file encryption) with data exfiltration and public extortion threats. The attackers maximize pressure by threatening both operational disruption AND data exposure.

b. CIA Triad Threats



ALL THREE OBJECTIVES COMPROMISED



Analysis Questions

Detailed Response

CONFIDENTIALITY

Sensitive data
stolen &
threatened for
release

INTEGRITY

Systems
encrypted
& data
potentially
corrupted

AVAILABILITY

Critical
systems locked
& inaccessible

c. Impact Analysis

OPERATIONS

Complete banking service
disruption, transaction
failures, branch closures

FINANCES

\$10M-\$100M+ costs,
fines, legal fees, customer
compensation

REPUTATION

Severe customer trust
erosion, media coverage,
brand damage

LEGAL RISK

Regulatory penalties,
lawsuits, compliance
violations

d. Early Detection



NETWORK ANOMALY MONITORING

Monitor for unusual data movement patterns, large file transfers to external locations, unauthorized database access, abnormal user behavior, and suspicious lateral network movement - these often precede extortion by days/weeks.

 Analysis Questions **Detailed Response****e. Parties Impacted**

 **INTERNAL:** All employees, executives, board members, IT/security teams

 **EXTERNAL:** Millions of retail/commercial customers, business partners, correspondent banks, regulators, shareholders, vendors, local communities, other financial institutions in payment networks

f. Third-Party Pressures

 **REGULATORY:** Federal banking regulators (OCC, FDIC, Fed), international regulators, data protection authorities requiring breach notifications

 **BUSINESS:** Payment networks (Visa, Mastercard), clearing houses, interbank systems, critical vendors, major corporate clients threatening to switch

 **PUBLIC:** Media scrutiny, social media pressure, political attention, competitor opportunism

Analysis Questions

Detailed Response

g. Social Media Crisis

IMMEDIATE ACTION REQUIRED!

 **CONTACT:** Reach employee directly for post removal

 **ASSESS:** Evaluate information disclosed and potential damage

 **RESPOND:** Issue official company statement, provide consistent messaging to all employees

 **FOLLOW-UP:** Implement social media guidelines, consider disciplinary action while maintaining support

h. Lock Out Adversary

1. NETWORK SEGMENTATION

Isolate affected systems, implement emergency partitioning

2. CREDENTIAL RESET

Force password changes, disable compromised accounts, enable MFA

3. ACCESS ELIMINATION

Close remote access vectors, patch vulnerabilities, deploy EDR tools

i. Postmortem Topic

 **INCIDENT TIMELINE & ATTACK VECTOR ANALYSIS**
Comprehensive review of initial access methods, network dwell time, attack progression, and missed detection

 Analysis Questions **Detailed Response**

opportunities. Critical for preventing future attacks and improving security controls.

 Preventive Measure**ZERO TRUST NETWORK ARCHITECTURE**

Why Critical: Traditional perimeter security failed. Zero Trust assumes no user/device is trustworthy, requiring continuous verification. Prevents lateral movement after compromise and limits attack blast radius through micro-segmentation and least-privilege access - essential for banking environments.